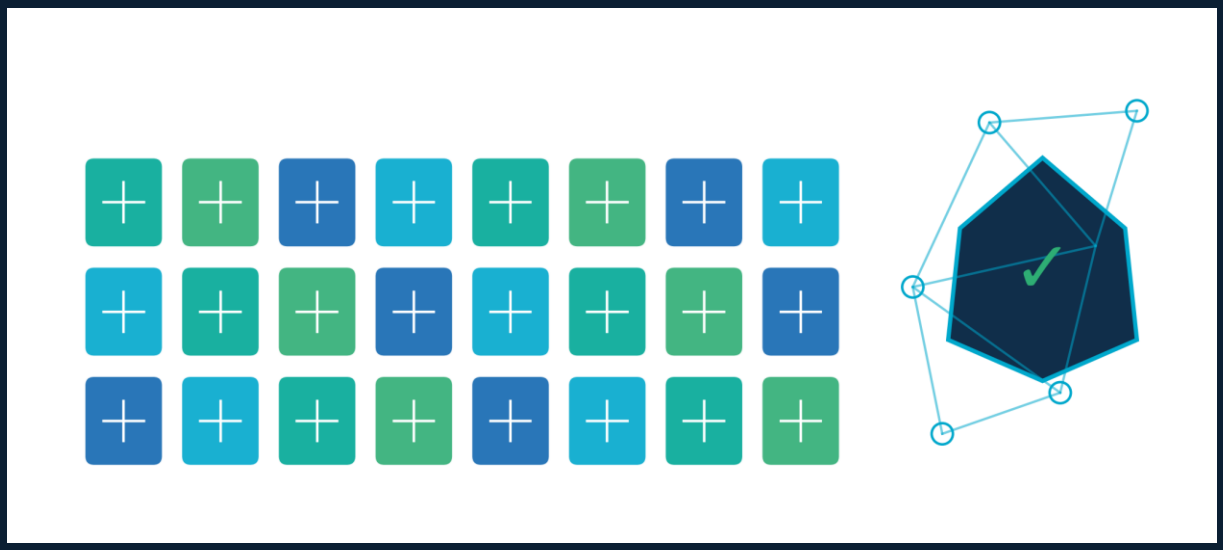
VGMarines

TrustSense-BMS

Battery Analytics Internship Project Handbook

Physics-Informed Virtual Sensors • Sensor Reduction • Automotive Cybersecurity • Blockchain Data Trust • LLM/RAG



Nine-month coordinated project for Computer Science / AI-ML interns

Prepared for: Interns | Version 1.0 | 24 July 2026

Document Control

Field	Details
Document title	TrustSense - BMS: Battery Analytics Internship Project Handbook
Organization	VG Marines
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Version / date	Version 1.0 / 24 July 2026
Status	Editable project definition and learning handbook

Important safety and scope statement

This handbook defines a research and demonstration project. It does not authorize removal of safety-critical sensors from a production EV battery, nor does it replace ISO 26262 functional-safety engineering, homologation, cell-manufacturer limits, or validation by qualified battery and vehicle engineers. Virtual sensor outputs must remain advisory or analytically redundant until independently verified and incorporated into an approved safety concept.

Executive Summary

The recommended umbrella project is TrustSense-BMS: a physics-informed battery analytics platform that reconstructs unmeasured cell or module temperatures, evaluates how many physical temperature sensors can be removed, detects sensor faults and cyber manipulation, and creates tamper-evident evidence of model and data provenance. The three interns work on separate but integrated work packages, sharing a common data model, test environment and demonstration dashboard.

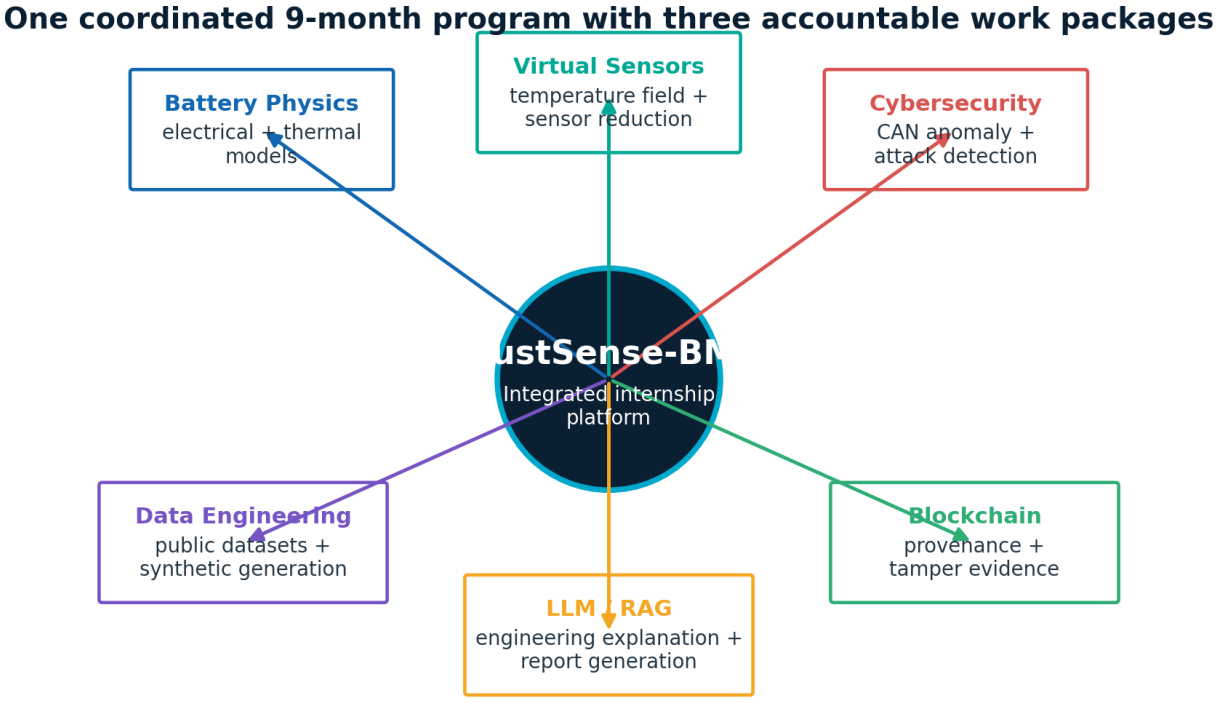


Figure 1. Core concept and relationships among the three intern work packages.

Intern	Primary ownership	Key technical output
Intern 1	Physics-informed virtual sensing and sensor placement	Full temperature-field reconstruction, hotspot estimation, uncertainty and sensor-ablation study
Intern 2	Battery sensor cybersecurity and CAN/CAN-FD intrusion detection	Fault/attack discrimination, real-time anomaly scoring and incident evidence
Intern 3	Trusted telemetry, blockchain and battery-domain LLM/RAG	Hyperledger Fabric provenance layer, smart contracts and engineering assistant

How to Use This Handbook

1. Study Part I before selecting algorithms. The interns should be able to explain voltage, current, SOC, resistance, heat generation and thermal gradients in their own words.
2. Use Part II to build models in increasing order of complexity. A deep model is accepted only after it beats transparent baselines on untouched data.
3. Use Parts III and IV to add cyber-resilience and provenance without allowing the LLM or blockchain components to become part of the real-time safety loop.
4. Use Parts V-VII as the formal project charter: ownership, datasets, monthly deliverables, tests, acceptance criteria and final handover package.

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Part III Cybersecurity and Adversarial Machine Learning

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Appendices Glossary, data schema, deliverable structure and references

PART I

Battery and BMS Foundations

The minimum electrical, electrochemical and thermal theory needed to build meaningful AI models

1. Why Battery Physics Matters to an AI Engineer

Battery data is not an ordinary collection of independent sensor channels. Voltage, current, temperature, SOC, cooling state and degradation are coupled by conservation laws and electrochemical constraints. A model that ignores these relationships can produce numerically accurate-looking predictions that are physically impossible or unsafe outside the training distribution.

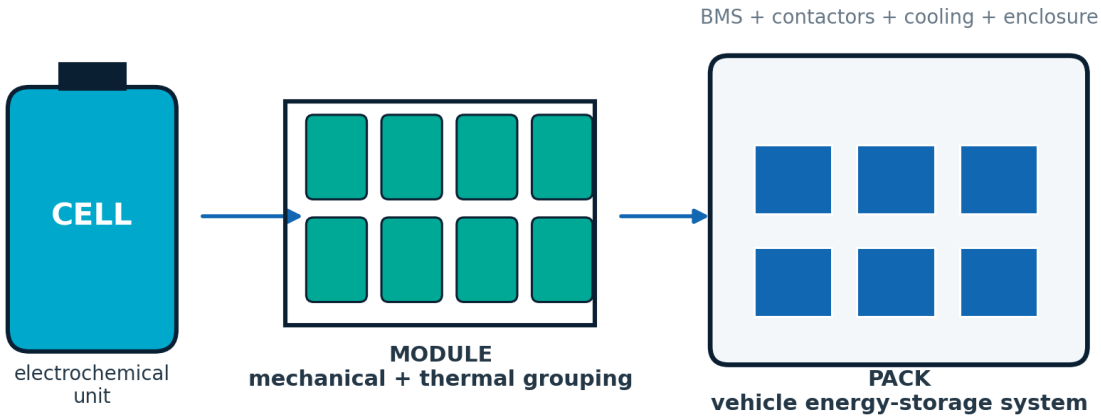
- **Current creates both useful electrochemical work and heat.** A rapid current increase can cause an immediate voltage drop and a delayed temperature rise.
- **Temperature changes battery behaviour.** Resistance, available power, charge acceptance, ageing rate and safety margins are temperature dependent.
- **SOC is hidden.** It is estimated from current integration, voltage behaviour and models; it is not measured directly by a single sensor.
- **Surface temperature is incomplete.** The core or an inaccessible cell can be hotter than the measured point, especially during high-rate operation or cooling faults.
- **A cyberattack can mimic physics.** A manipulated sensor value may resemble drift, calibration error, a real hotspot or an unexpected operating condition.

Learning objective

At the end of Part I, each person should be able to draw the battery hierarchy, explain the sign and units of every input, derive a simple voltage and thermal model, and identify which quantities are measured, estimated or commanded.

2. Battery System Hierarchy

Battery hierarchy: measurements originate at different physical levels



F

igure 2. Cell, module and pack hierarchy.

Level	Typical content	Examples of relevant measurements
Cell	Electrodes, separator, electrolyte, current collectors and safety devices	Cell voltage, selected surface temperatures, impedance in laboratory tests
Module	Mechanically and thermally grouped cells, busbars and local sensing	Module voltage, several temperatures, cooling contact conditions
Pack	Modules, BMS, contactors, current sensor, isolation monitor, cooling system and enclosure	Pack current/voltage, cell voltages, temperatures, coolant temperatures, contactor and isolation status
Vehicle / charger	VCU, thermal controller, inverter, charger, cloud and service tools	Torque/power demand, charging request, ambient conditions, diagnostics and communication data

3. Simplified Lithium-Ion Cell Physics

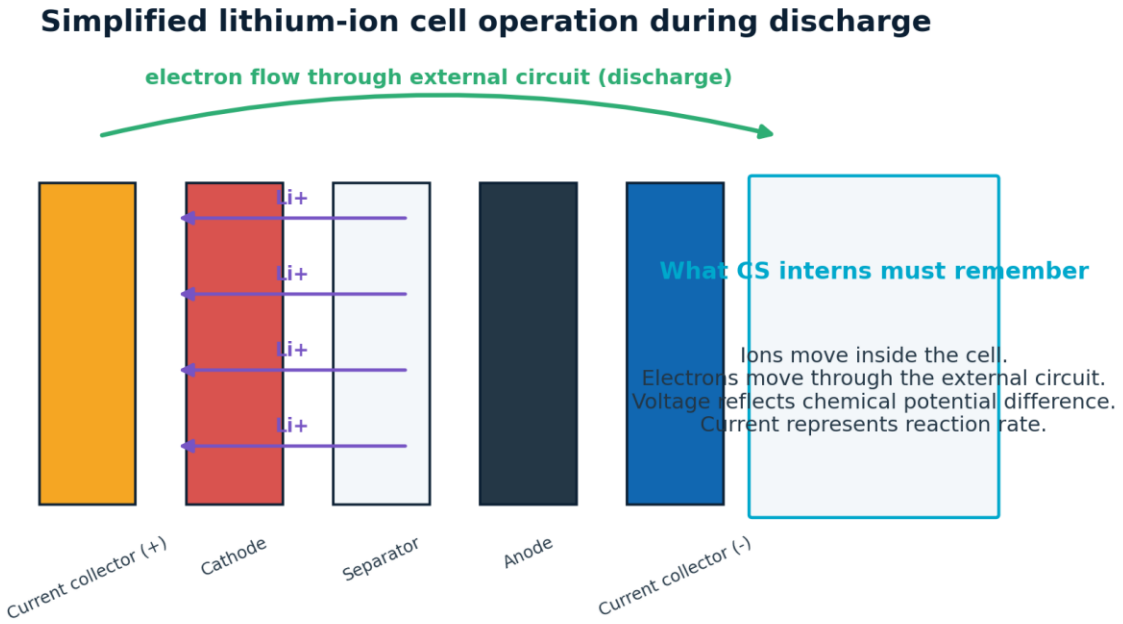


Figure 3. Simplified movement of lithium ions and electrons during discharge.

A lithium-ion cell stores energy by maintaining lithium in different chemical states at the two electrodes. During discharge, lithium ions travel through the electrolyte and separator while electrons travel through the external circuit. During charge, both flows reverse. The separator allows ionic transport while preventing direct electronic contact between electrodes.

Term	Practical meaning for the project
Open-circuit voltage (OCV)	Equilibrium-like cell voltage after sufficient rest; strongly related to chemistry, SOC and temperature.
Overpotential / polarization	Additional voltage loss caused by reaction kinetics and mass transport under current.
Internal resistance	A convenient aggregated quantity that creates immediate voltage drop and irreversible heat; it varies with SOC, temperature and ageing.
Capacity (Ah)	Charge that can be removed under defined conditions. One ampere-hour equals 3600 coulombs.
Energy (Wh)	Integral of voltage multiplied by current over time; approximately nominal voltage multiplied by capacity for rough estimates.

4. Essential Electrical Quantities and Sign Conventions

Power: $P(t) = V(t) \times I(t)$

Define once whether positive current means discharge or charge; use the same convention in every dataset and model.

Energy: $E = \int [V(t) \times I(t)\, dt]$

With V in volts, I in amperes and time in hours, E is in watt-hours.

C-rate: $C_{rate} = |I| / C_{nominal}$

For a 50 Ah cell, 1C is 50 A and 0.5C is 25 A.

Coulomb-counted SOC: $SOC(t) = SOC(t0) - (\eta / C_{available}) \int [I(t)\, dt]$

The sign depends on the adopted current convention; eta represents charge/discharge efficiency.

Quantity	Unit	Measured or estimated?	Common data issue
Cell voltage	V	Measured	offset, wiring drop, sampling skew, impossible step
Pack current	A	Measured	offset, polarity error, saturation, dynamic lag
Temperature	°C or K	Measured at selected points	bias, drift, contact error, thermal lag, dropout
SOC	%	Estimated	initialization error, current-sensor bias, chemistry-dependent OCV
SOH	%	Estimated	definition inconsistency: capacity SOH vs resistance/power SOH
Cooling state	command / flow / speed	Measured or commanded	command does not prove physical flow

5. SOC, SOH and RUL

State of charge describes the present charge inventory relative to usable capacity. State of health describes degradation relative to a reference state, but organizations must specify the definition: remaining capacity, resistance increase, available power or a composite indicator. Remaining useful life predicts time or cycles until a defined end-of-life criterion is reached.

Capacity SOH = $C_{available} / C_{reference} \times 100\%$

The reference capacity and test conditions must be documented.

Resistance SOH indicator = $R_{reference} / R_{current} \times 100\%$

Alternative formulations are possible; never mix definitions without conversion.

Why this matters for virtual temperature sensing

The same current can produce different heat in a new cell and an aged cell because resistance and reaction losses change. SOH, recent history and learned resistance features therefore help the model generalize across life.

6. Chemistry-Dependent Voltage Behaviour

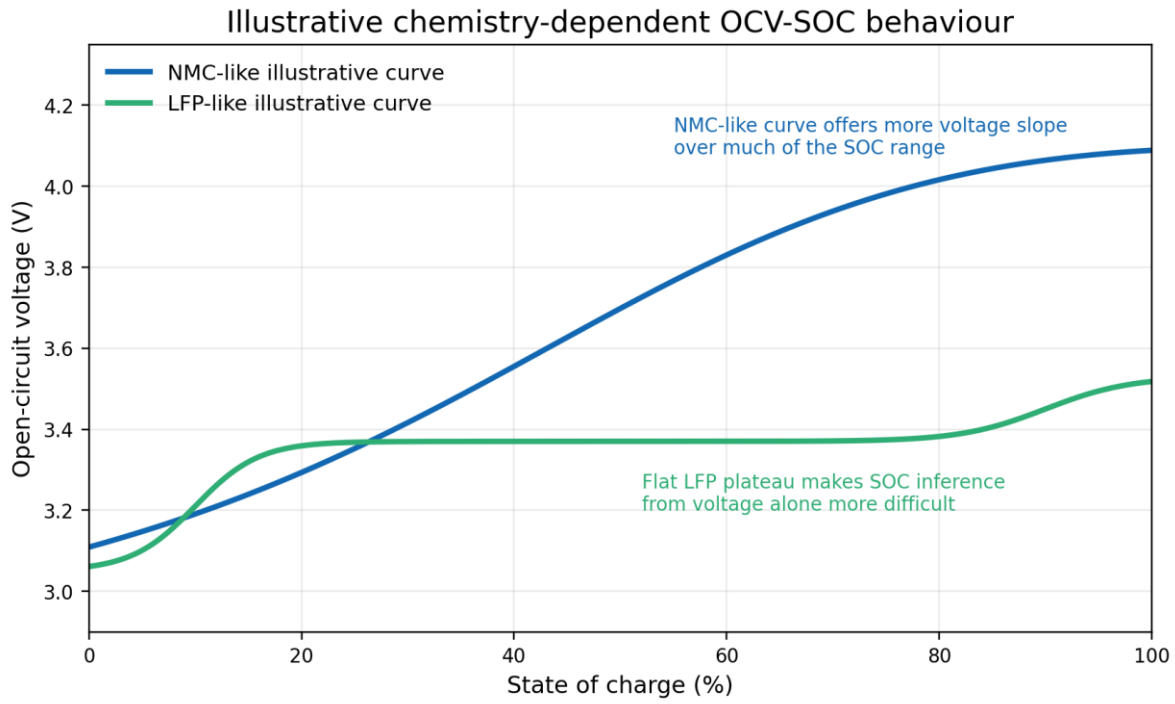


Figure 4. Illustrative OCV-SOC curves showing why chemistry must be a model input or separate calibration.

LFP commonly exhibits a broad voltage plateau, so a small voltage error can correspond to a large SOC uncertainty in the middle of the range. NMC-like chemistries often provide more voltage slope over a wider range. Temperature, hysteresis, current history and ageing further change observed terminal voltage. Consequently, a model trained on one chemistry cannot be assumed valid for another without transfer testing and recalibration.

7. Equivalent-Circuit Models

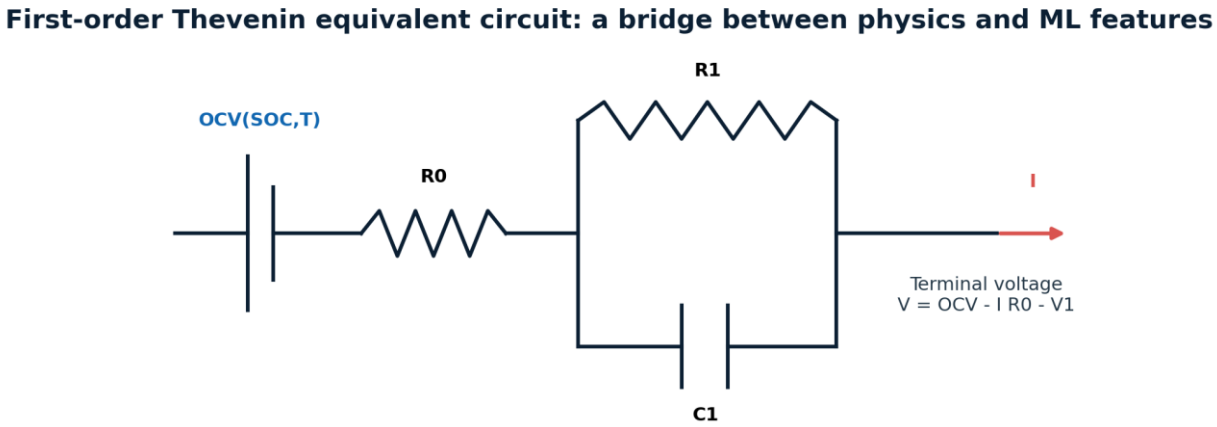


Figure 5. First-order Thevenin equivalent-circuit model.

Equivalent-circuit models replace detailed electrochemistry with a small set of dynamic electrical elements. They are fast enough for BMS use and provide interpretable features for AI. A first-order Thevenin model uses an OCV source, an ohmic resistance R0 and a polarization branch R1-C1.

Terminal voltage: $V = OCV(SOC,T) - I R0 - V1$

Polarization state: $dV1/dt = -(1 / (R1 C1)) V1 + I / C1$

- **R0 feature:** captures instantaneous voltage drop and contributes approximately I²R0 heat.
- **R1-C1 feature:** captures slower electrochemical polarization and relaxation.
- **Residual feature:** measured voltage minus model voltage can reveal sensor error, parameter drift or abnormal behaviour.

8. Battery Heat Generation and Thermal Dynamics

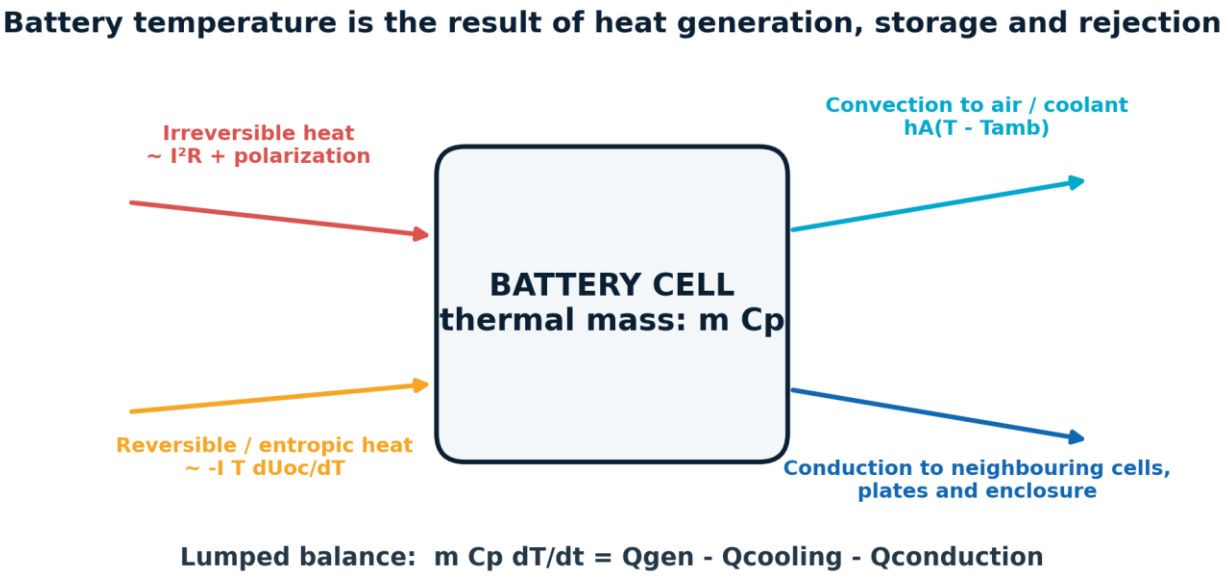


Figure 6. Simplified cell heat-generation and heat-rejection balance.

Battery temperature changes because heat is generated internally, stored in the cell thermal mass, conducted to neighbouring structures and removed by air or coolant. A commonly used heat expression separates irreversible and reversible contributions. For an internship model, the interns may start with I²R and then add an entropic term if suitable parameters are available.

Approximate irreversible heat: $Q_{irr} = I^2 R_{internal}$

General electro-thermal form: $Q_{gen} = I (V - U_{oc}) - I T (dU_{oc}/dT)$

The exact sign depends on current convention and equation formulation; verify against the chosen reference model.

Lumped thermal model: $m C_p dT/dt = Q_{gen} - hA(T - T_{ambient}) - Q_{conduction}$

Symbol	Meaning	How interns may obtain it
m	Cell mass	Cell data sheet or measured value
Cp	Specific heat capacity	Literature, parameter set or calibration
hA	Effective convective cooling coefficient times area	Identification from experiments or simulation
R_internal	Effective resistance	Pulse test, ECM identification or learned feature
dUoc/dT	Entropic coefficient	Chemistry/SOC-dependent laboratory data or literature parameter set

9. Why Temperature Is Spatial

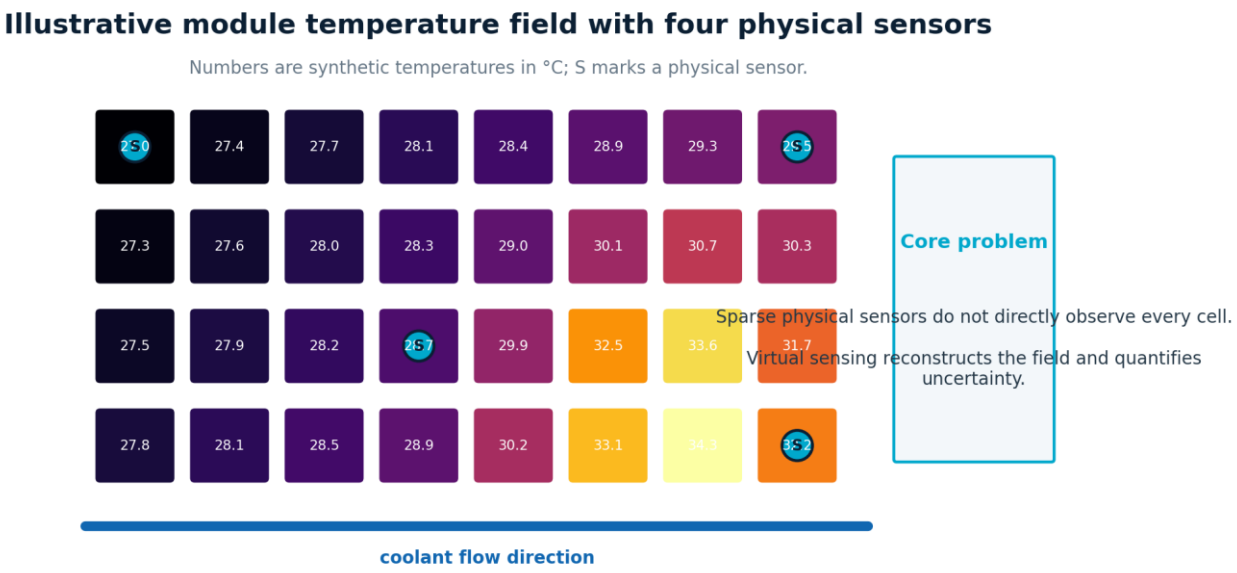


Figure 7. Synthetic module temperature field with sparse physical sensors.

Cells are not thermally identical. Busbar resistance, cooling-path position, airflow, coolant temperature rise, cell ageing, contact resistance, enclosure geometry and neighbouring cells create nonuniform temperatures. A point sensor measures its local contact location with its own response time; it does not guarantee the maximum internal or surface temperature is observed.

- **Hotspot:** the maximum temperature or high-risk location at a given time.
- **Temperature gradient:** difference between hottest and coldest cells or between core and surface.
- **Thermal lag:** delay between heat generation and measured sensor response.
- **Boundary condition:** ambient, coolant or contact condition that determines heat rejection.
    - **Thermal coupling:** heat flow between neighbouring cells and structures, motivating graph-based models.

10. BMS Sensors and Safety Functions

Sensor / signal	Purpose	Virtualization suitability in this project
Cell voltage	Over/undervoltage protection, balancing, SOC and diagnostics	Use virtual estimates for analytical redundancy only; do not remove required production sensing.
Pack current	Power/energy calculation, SOC integration, overcurrent protection	Do not replace as the sole protection measurement in the internship project.
Cell/module temperature	Overtemperature protection, cooling and diagnostics	Primary sensor-reduction research target, with conservative fallback and uncertainty.
Coolant inlet/outlet temperature	Thermal-control supervision and model boundary condition	May be estimated for diagnostics, but physical cooling verification remains important.
Isolation resistance	High-voltage electrical safety	Outside the removal scope.
Contactor feedback	Confirms high-voltage switching state	Outside the removal scope.

Recommended first research boundary

Concentrate sensor reduction on redundant module or surface temperature sensing. Treat cell voltage, pack current, isolation monitoring, contactor feedback and independent overtemperature protection as non-removable safety channels in the prototype.

11. Sensor Error and Failure Physics

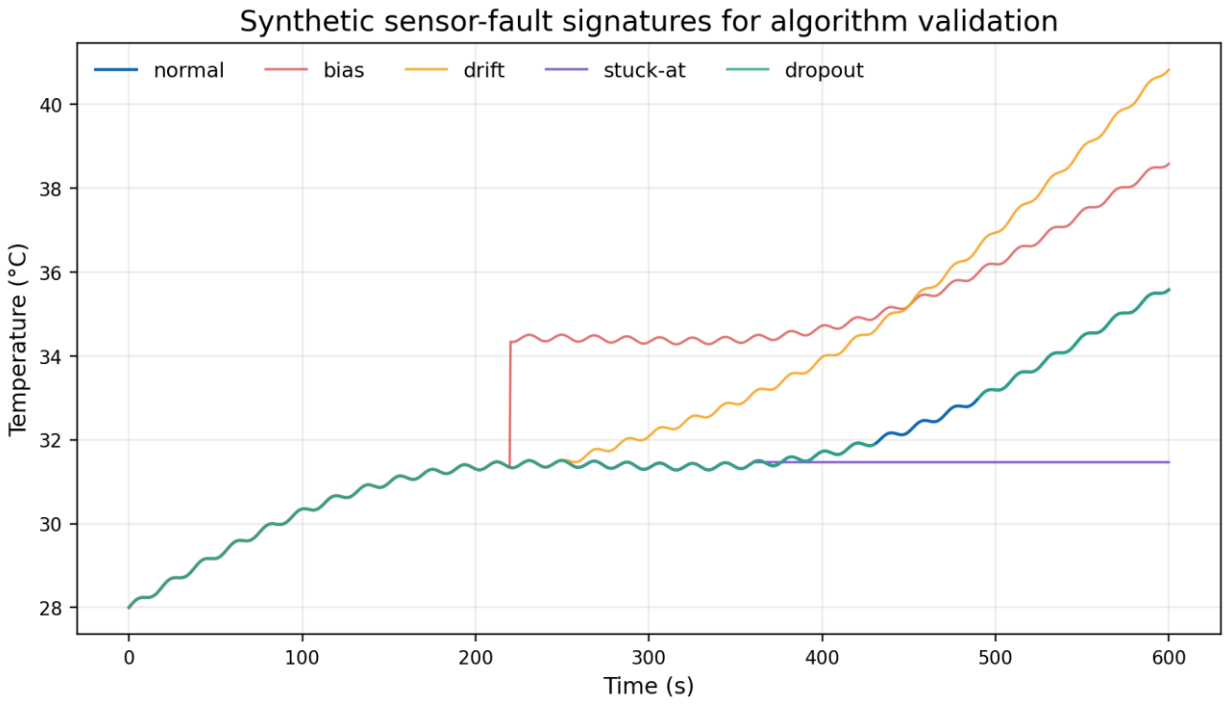


Figure 8. Synthetic examples of sensor bias, drift, stuck-at and dropout faults.

Failure mode	Signal signature	Potential physical confusion
Bias / offset	Constant difference from expected value	Real local gradient or poor sensor contact
Drift	Slowly increasing/decreasing residual	Ageing-related heat increase or changing cooling
Stuck-at	Value remains constant despite dynamic operation	Long rest or very slow thermal response
Dropout	Missing / invalid samples	Communication interruption or power issue
Noise increase	Higher variance or spikes	Switching interference, loose contact or real transient
Time shift	Correct shape but delayed/advanced	Clock synchronization or thermal lag

PART II

Sensor Virtualization and Sensor Reduction

From transparent baselines to physics-informed graph models and safe uncertainty handling

12. Definition and Objective

A virtual sensor is a software estimator that reconstructs a quantity that is unavailable, unmeasured or considered unreliable, using other measurements and a model. In this project, the primary target is the temperature of unsensed cells or locations. Sensor reduction asks a second question: what is the smallest physical sensor set that still achieves required error, hotspot detection, fault isolation and safety coverage?

Virtual temperature vector: T_hat(t) = f(I, V_cells, T_sparse, SOC, SOH, cooling, position, history)

Sensor-placement objective: minimize |S| subject to error <= limit, hotspot recall >= target, and fault isolation requirements

S is the selected set of physical sensor locations.

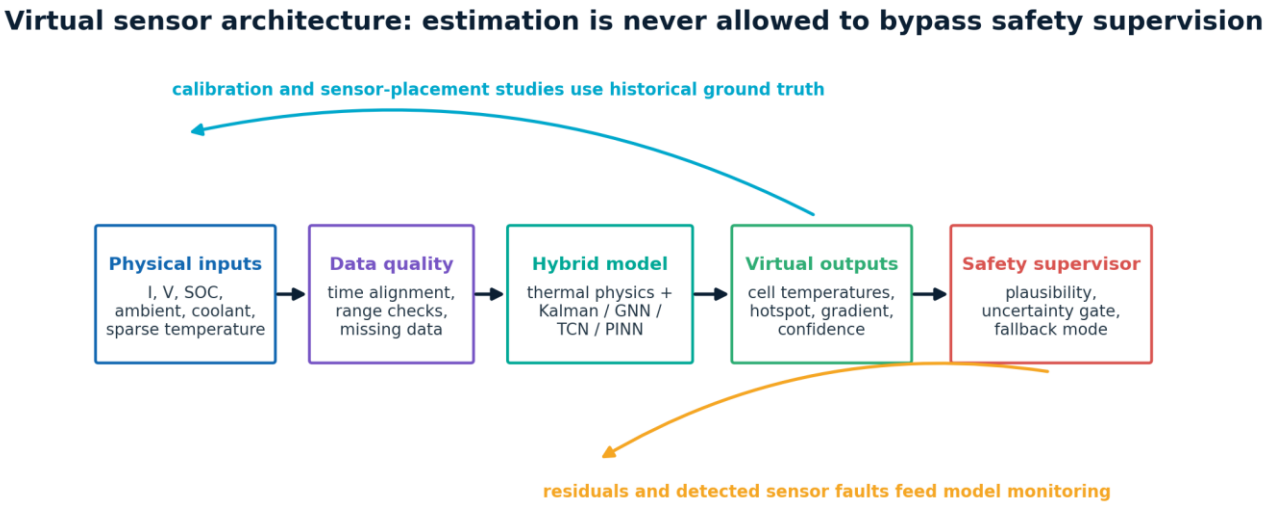


Figure 9. End-to-end virtual sensor workflow with a mandatory safety supervisor.

13. Observability and Analytical Redundancy

Observability describes whether hidden states can be inferred from available inputs and outputs. Analytical redundancy means the same physical behaviour is represented through multiple relationships rather than duplicate hardware. For example, pack current, voltage response and a thermal model can predict a plausible temperature trend; disagreement with a physical sensor creates a residual for fault detection.

- **Not observable:** two different thermal states generate nearly identical available measurements.
- **Weakly observable:** the state can be inferred, but noise or model uncertainty causes large error.
- **Analytically redundant:** a measurement can be checked against a model or neighbouring measurements.

- **Fault isolable:** the system can identify which sensor or cell is likely responsible, not merely that an anomaly exists.

Key design rule

Sensor reduction is not only a prediction problem. The selected sensor set must also support fault detection and isolation. A configuration with low average RMSE may still be unacceptable if it misses the hottest cell or cannot distinguish a failed sensor from a real hotspot.

14. Model Development Ladder

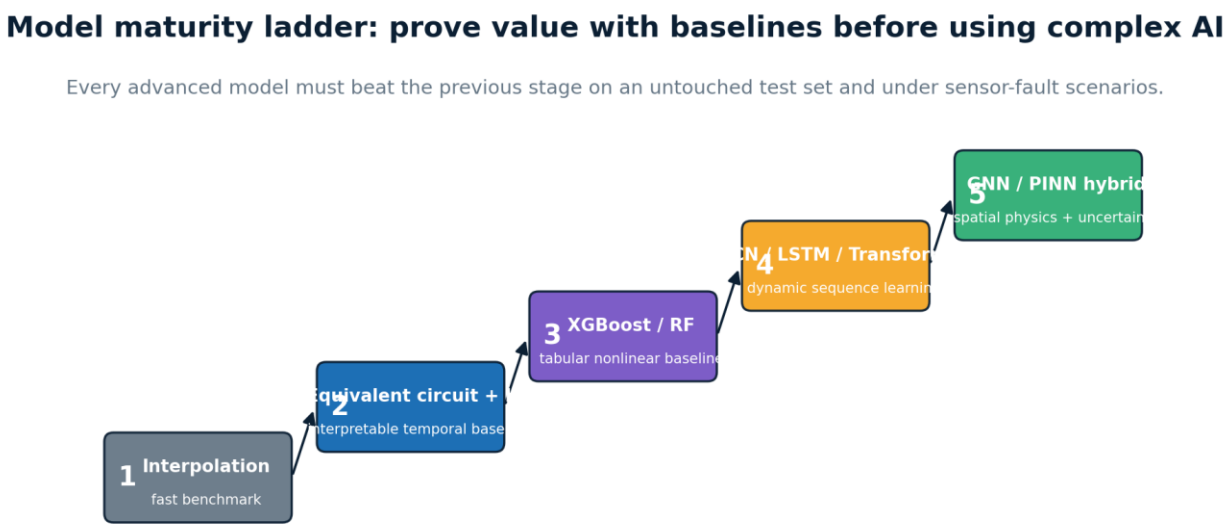


Figure 10. Recommended progression from simple baselines to hybrid AI.

Model family	Strength	Limitation	Required baseline test
Interpolation / nearest neighbour	Very fast and transparent	Cannot capture dynamic heating or irregular geometry	Compare error versus sensor spacing and hotspot location
Thermal RC model + Kalman filter	Interpretable, uncertainty-capable, data efficient	Requires parameter identification and simplified geometry	Validate step response and residual whiteness
Random Forest / XGBoost	Strong for engineered tabular features	Limited temporal/spatial memory unless features are designed	Cell-disjoint and cycle-disjoint test
TCN / LSTM / Transformer	Learns dynamic history	Can overfit and extrapolate poorly	Unseen drive-cycle and temperature tests
GNN	Represents cell adjacency and irregular layouts	Graph definition and transfer to new pack geometry require care	Layout transfer and missing-node tests
PINN / hybrid residual model	Uses conservation laws and can improve data efficiency	Training and loss balancing can be difficult	Physics-residual and out-of-range tests

15. Graph Neural Networks for Battery Modules

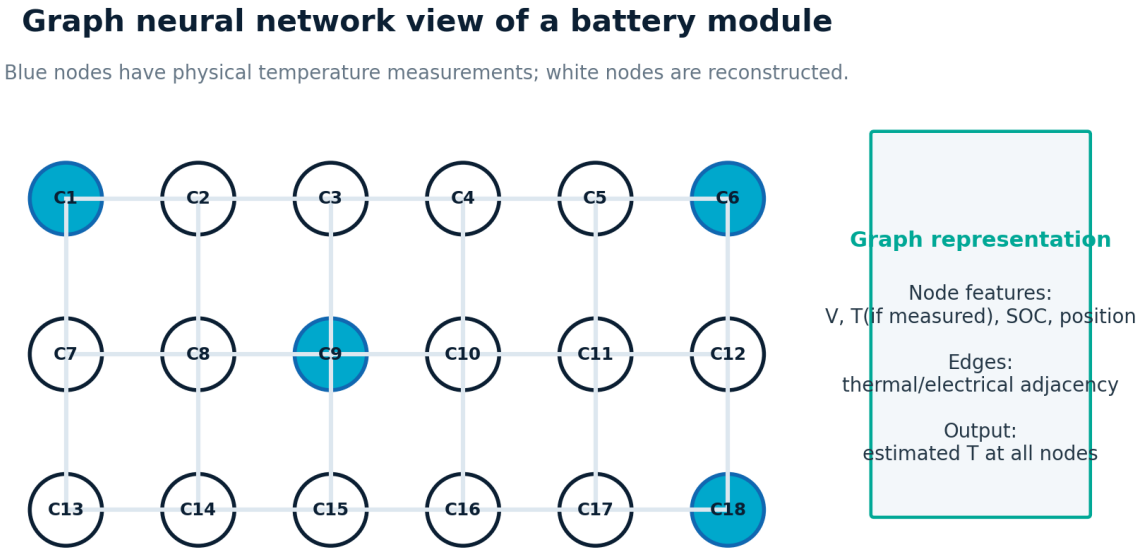


Figure 11. Battery cells represented as graph nodes with thermal/electrical adjacency edges.

A battery module is naturally represented as a graph. Each cell is a node containing voltage, measured temperature if available, SOC, position and cooling features. Edges connect thermally adjacent cells, electrically related cells or cells sharing a cooling plate. Message passing allows information from sensed nodes and operating conditions to reconstruct unsensed nodes.

GNN layer: h_i^(k+1) = sigma(W_self h_i^k + aggregate_j in N(i) [W_neigh h_j^k + edge_features_ij])

- **Node mask:** indicates whether a physical temperature is available, preventing the network from confusing a missing sensor with a true zero.
- **Position encoding:** row, column, module, distance to cooling inlet and boundary location.
- **Edge features:** distance, shared plate, contact conductance or coolant-flow direction.
- **Output head:** mean temperature estimate plus uncertainty or quantiles.

16. Physics-Informed and Hybrid Modelling

Physics-informed learning adds penalties or model structure so predictions respect thermal dynamics. A practical alternative to a pure PINN is a hybrid residual model: a simple thermal model predicts the main trend and a neural network learns the remaining error. This is often easier to train and explain than solving full electrochemical partial differential equations inside the loss.

Hybrid estimate: T_hat = T_physics + DeltaT_ML

Example loss: L = w_data L_data + w_phys L_energy_balance + w_smooth L_spatial + w_unc L_calibration

Loss term	Purpose	Example
Data loss	Fit measured/ground-truth temperatures	MAE or Huber loss
Energy-balance	Penalize violation of heat	Residual of lumped thermal ODE

Loss term	Purpose	Example
loss	storage/generation/rejection	
Spatial smoothness	Encourage physically plausible neighbouring temperatures	Graph Laplacian penalty
Hotspot loss	Prioritize maximum-temperature location and value	Weighted top-k or ranking loss
Uncertainty calibration	Align predicted confidence with actual errors	Negative log likelihood or quantile loss

17. Sensor Placement and Reduction Algorithms

The interns should compare at least four placement strategies. The best strategy is selected on validation data and then frozen before final testing. Random placement is necessary as a control because an optimized method must demonstrate a repeatable advantage.

Strategy	Method	Why include it
Uniform / engineering layout	Corners, inlet/outlet, centre and expected hotspot	Simple, explainable baseline
Random subsets	Repeated Monte Carlo selection	Quantifies chance performance and confidence interval
Greedy error reduction	Add the sensor that most reduces validation error	Practical optimization baseline
PCA/POD / QR pivoting	Select locations that capture dominant spatial modes	Data-efficient linear observability method
Mutual information / uncertainty	Select location with highest information gain	Targets uncertainty reduction
Graph/gradient optimization	Learn or search a placement mask with constraints	Advanced method for irregular modules

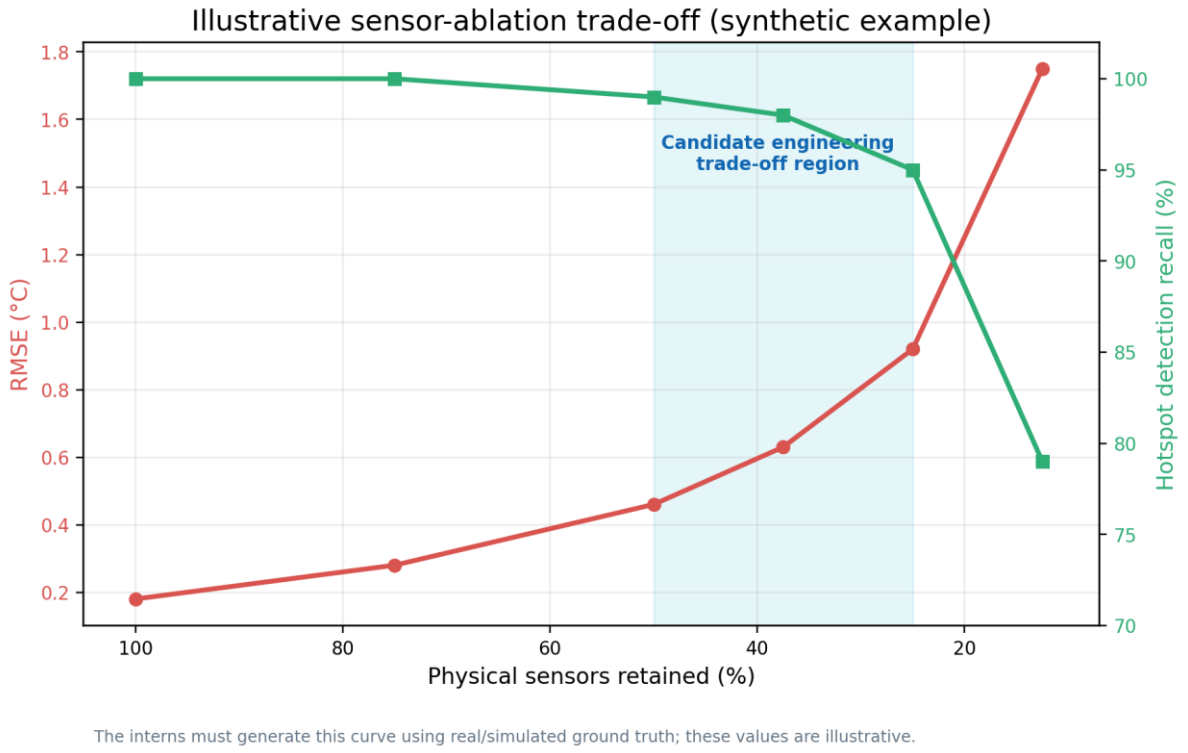


Figure 12. Synthetic example of the sensor-count versus performance trade-off.

18. Uncertainty, Confidence and Safe Fallback

A virtual sensor should not output only a single number. It should quantify confidence and refuse to substitute for a sensor when operating conditions are outside training coverage, residuals are inconsistent or uncertainty is high. The safety supervisor then falls back to conservative operation or marks the signal unavailable for the demonstration.

Mechanism	Implementation option	Acceptance evidence
Prediction interval	Ensemble variance, quantile regression or probabilistic head	Coverage close to stated confidence on untouched tests
Out-of-distribution detector	Feature distance, density model or deep ensemble disagreement	Correctly flags unseen chemistry/range scenarios
Residual monitor	Measured minus estimated temperature/voltage	Fault detection delay and false alarms per hour
Fallback rule	Use physical sensor only, conservative maximum, or unavailable flag	Tested response for missing model, data or network
Model-health monitor	Latency, checksum, input quality and version	Evidence in dashboard and blockchain event log

PART III

Cybersecurity and Adversarial Machine Learning

Protecting battery sensor data, in-vehicle communication and the AI development lifecycle

19. Battery Analytics Threat Model

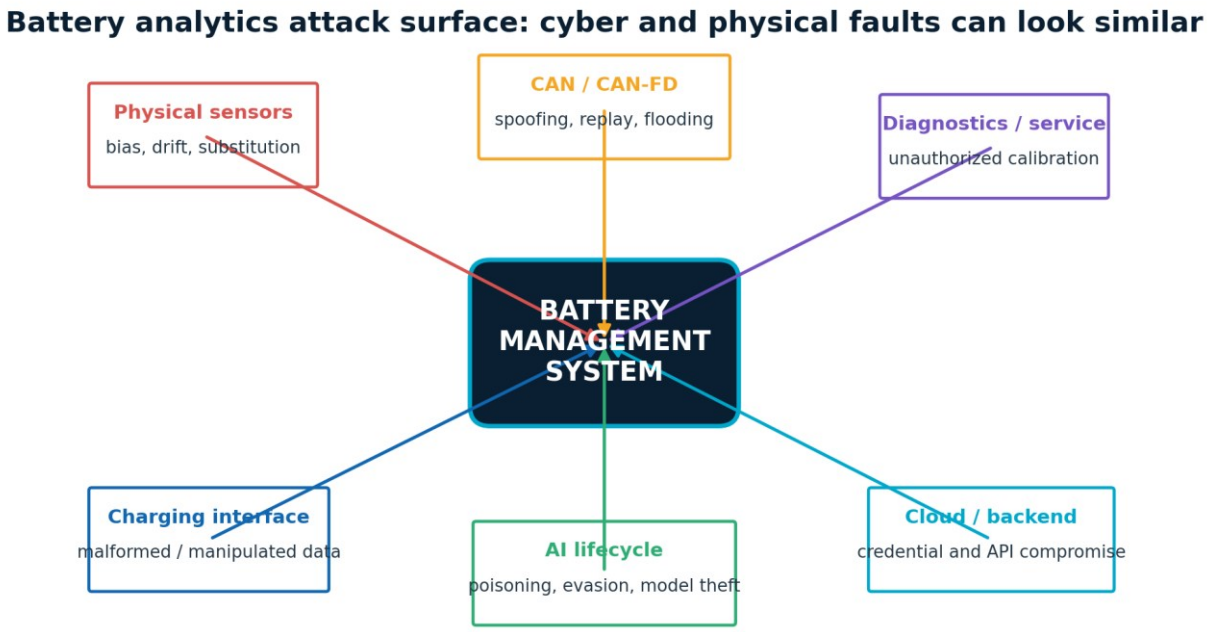


Figure 13. Battery analytics attack surface.

Battery analytics crosses physical sensors, in-vehicle networks, service tools, charging interfaces, cloud APIs and AI pipelines. The intern must distinguish accidental failure from deliberate manipulation, but the first-stage detector can treat both as an integrity anomaly and then perform root-cause classification.

Asset	Threat example	Potential consequence
Temperature sensor data	Bias or replay to hide overheating	Delayed cooling or missed warning
Cell voltage data	Spoofed values or frozen frame	Incorrect balancing/SOC inference or false normal state
CAN/CAN-FD timing	Flooding, injection or impersonation	Communication delay, ECU confusion or false state
Calibration / model file	Unauthorized threshold or parameter change	Systematic unsafe estimates
Training dataset	Poisoned labels or backdoor trigger	Model behaves incorrectly under selected condition
LLM knowledge base	Prompt injection or untrusted document	Misleading incident explanation or data leakage
Provenance record	Event deletion or modification	Loss of auditability and battery-passport trust

20. CAN/CAN-FD Basics for the Project

Controller Area Network is a broadcast bus widely used by vehicle electronic control units. Classical CAN data frames identify message type through an arbitration ID but do not natively provide source authentication or confidentiality.

Lower numerical arbitration identifiers have higher priority, so high-rate dominant IDs can disrupt access to the bus. CAN-FD increases payload and data-rate capability but still requires an automotive security architecture around the protocol.

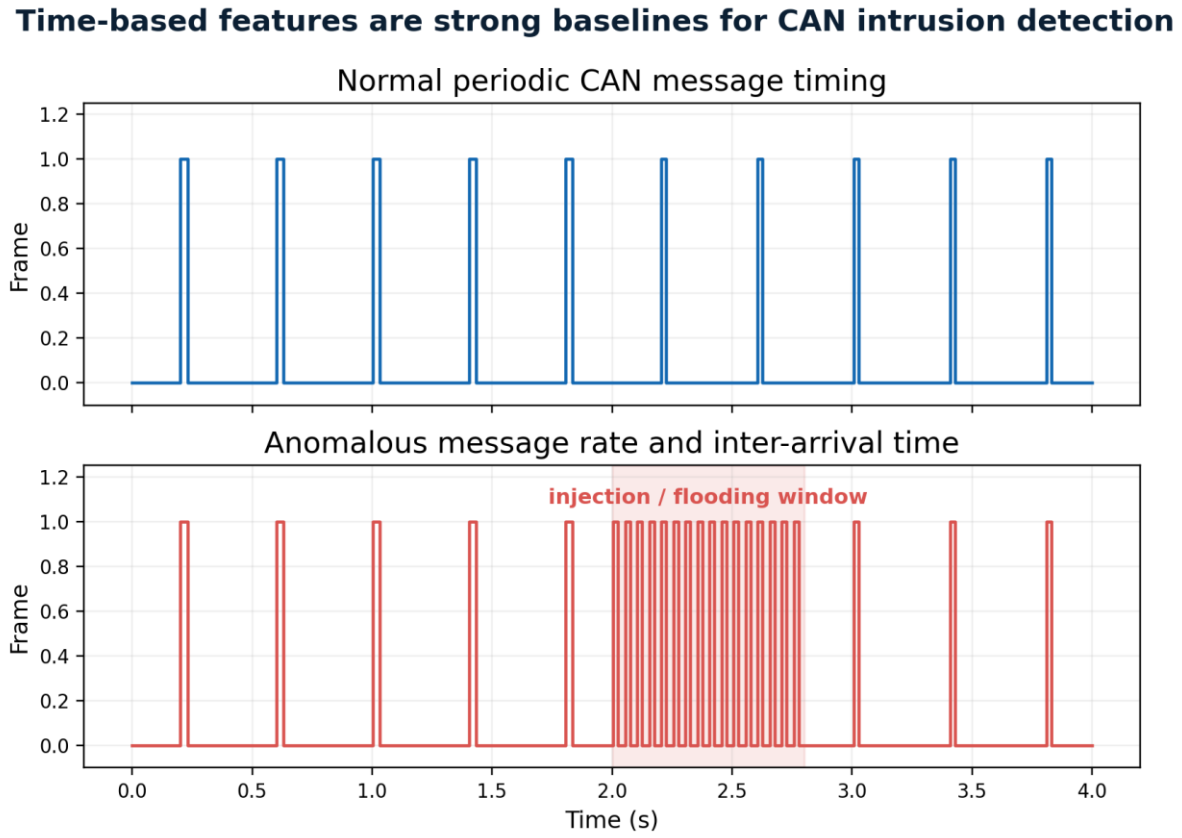


Figure 14. Normal and anomalous CAN message timing.

Feature group	Examples
Timing	inter-arrival time, message frequency, jitter, burst length, missing-period count
Identifier	CAN ID distribution, new/rare ID, priority pattern
Payload	byte values, bit transitions, entropy, decoded signal rate
Sequence/context	previous IDs, window embeddings, correlation with current/voltage/temperature
Physical consistency	temperature cannot jump instantly; pack voltage must be consistent with cells; SOC and current direction must agree

21. Attack and Fault Classes to Implement

Class	Injection method for safe research	Expected detector evidence
Stuck-at / frozen frame	Hold one value or replay an old window in software	low variation, timestamp/payload repetition, model residual
Bias	Add constant offset to selected sensor	persistent residual and neighbour inconsistency
Drift	Add gradually increasing offset	trend in residual, change-point evidence
Dropout	Remove packets or replace with invalid marker	missing-rate alarm and fallback
Flooding / DoS	Replay public CAN attack data or isolated virtual CAN test	abnormal inter-arrival time and bus load
Fuzzing	Use public labelled data; do not inject into a road vehicle	rare IDs/payloads, entropy and sequence anomaly
Spoofing	Replay or alter a known ID in an offline/virtual environment	timing conflict, payload inconsistency, physical residual
Stealth manipulation	Slow bias within normal value range	physics-based residual and multivariate detector

22. Intrusion-Detection Model Ladder

Stage	Model	Purpose
A	Rules and statistical thresholds	Understand message rates, ranges and simple physical constraints
B	Isolation Forest / One-Class SVM	Unsupervised anomaly baseline trained mainly on normal data
C	Autoencoder / LSTM autoencoder	Learn multivariate temporal reconstruction error
D	TCN / Transformer classifier	Learn labelled attack windows and timing context
E	Graph or multimodal detector	Combine CAN IDs, decoded signals, battery physics and sensor residuals
F	Open-set / uncertainty layer	Reject unseen attacks and avoid overconfident classification

Anomaly score example: $A = w1 \times reconstruction_error + w2 \times physics_residual + w3 \times timing_score$

23. Fault-versus-Attack Discrimination

A real sensor fault and a cyberattack can produce the same numerical value. Classification therefore uses context: whether the sensor hardware diagnostic changed, whether CAN timing is abnormal, whether multiple copies of an ID conflict, whether the signal violates thermal dynamics, and whether a service or network event occurred.

Evidence	More suggestive of physical fault	More suggestive of cyberattack
Sensor electrical diagnostic	open/short or ADC error	often normal unless attacker also manipulates diagnostic
CAN timing	normal periodic transmission	duplicate frames, burst, unusual source timing
Neighbouring sensors	local physical pattern possible	inconsistent or impossible pattern
Thermal model residual	may grow with sensor contact problem	may be abrupt, coordinated or timed to an event
Authentication / service log	no unauthorized action	credential, diagnostic session or firmware anomaly

24. Adversarial ML and LLM Security

The AI system itself is an attack surface. Training-time poisoning can alter model behaviour; inference-time evasion can manipulate input features; model extraction and membership inference can expose intellectual property or data; and a retrieval-augmented LLM can be influenced by untrusted documents or prompts. The project should use the NIST adversarial-ML taxonomy as a common vocabulary and maintain a threat register for both predictive and generative models.

- **Dataset controls:** hashes, immutable raw zone, review of labels, lineage and train/test separation.
- **Model controls:** signed model artifact, checksum at load, versioned model card and rollback.
- **Inference controls:** range checks, OOD detection, rate limits and physics plausibility.
- **RAG controls:** approved-source allowlist, role-based retrieval, prompt-injection filtering and citations.
- **Human control:** LLM output is advisory and requires review before any maintenance or safety decision.

25. Standards-Aligned Cybersecurity Process

ISO/SAE 21434 defines lifecycle cybersecurity engineering requirements for road-vehicle electrical/electronic systems. NISTIR 8473 provides a risk-oriented cybersecurity profile for the EV and extreme-fast-charging ecosystem. The internship will not claim formal compliance, but it should adopt the discipline of asset identification, threat analysis, security requirements, verification, incident evidence and lifecycle monitoring.

Process step	Internship artifact

Process step	Internship artifact
Asset and interface inventory	BMS data-flow and attack-surface diagram
Threat scenario definition	Threat register with precondition, attack path and impact
Risk treatment	Security requirement and mitigation mapping
Verification	Repeatable offline/virtual attack tests with pass/fail criteria
Monitoring	Anomaly dashboard, event severity and provenance log
Incident response	LLM-assisted draft report plus human-approved final report

PART IV

Blockchain and Trusted Battery Data

Using a permissioned ledger for provenance, governance and battery-passport evidence

26. What Blockchain Adds - and What It Does Not

Blockchain is useful when several organizations need a shared, tamper-evident history without allowing one party to silently rewrite records. It does not make inaccurate sensor data correct, does not replace cybersecurity controls, and is not suitable for storing every high-frequency sample. The project therefore stores bulk telemetry off-chain and records hashes, model versions, approvals and lifecycle events on-chain.

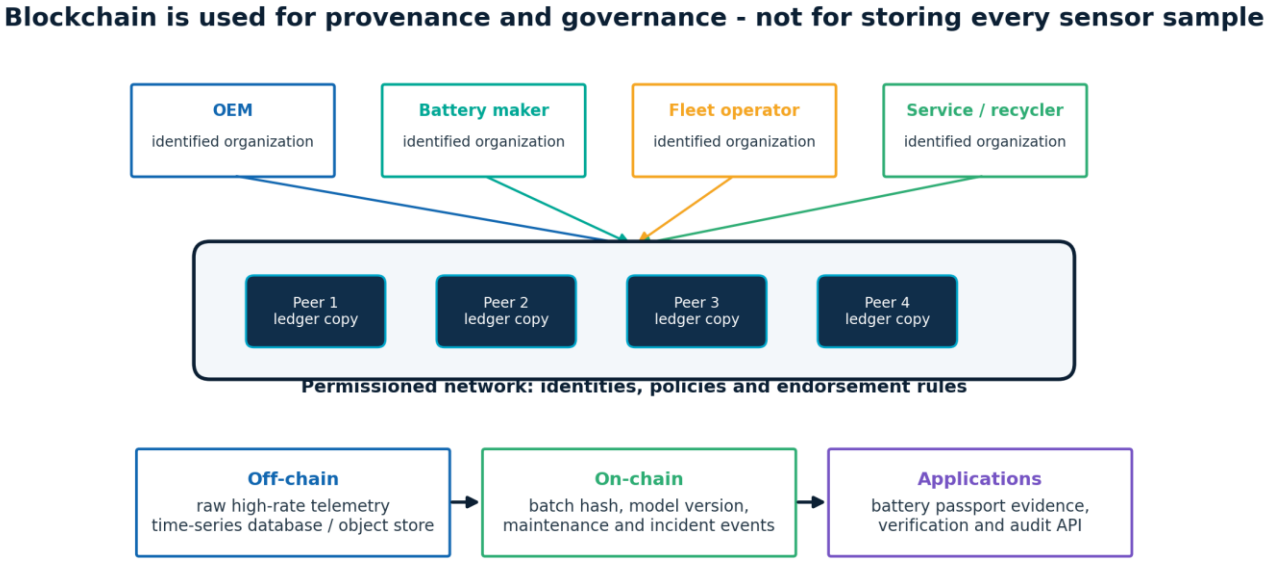


Figure 15. Permissioned blockchain architecture with off-chain telemetry.

Store off-chain	Record on-chain
Raw CAN logs and high-frequency sensor samples	Cryptographic hash of a data batch and time range
Large model files and training artifacts	Model identifier, checksum, version and approval status
Thermal maps and detailed test results	Test-result hash, status and responsible identity
Private service documents	Authorized event reference or private-data collection entry
LLM context documents	Document hash, access classification and revision

27. Why Hyperledger Fabric Is Suitable

Hyperledger Fabric is an enterprise-grade permissioned distributed-ledger platform. Participants have verifiable identities, policies control access and governance, and endorsement policies define which organizations must approve a transaction. These characteristics fit a consortium of an OEM, battery manufacturer, fleet operator, service organization and recycler better than an anonymous public blockchain.

Fabric concept	Project mapping
Organization / MSP	OEM, battery maker, fleet, VG Marine/service and recycler identities
Peer	Organization node maintaining ledger and executing smart-contract transactions

Fabric concept	Project mapping
Channel	Controlled collaboration domain for selected participants
Chaincode / smart contract	Battery registration, model approval, telemetry hash and incident event logic
Endorsement policy	Required organizations that approve selected events
Private data collection	Sensitive commercial data shared only with authorized organizations while a hash remains on the ledger

28. Proposed Smart Contracts

Contract function	Minimum fields	Validation rule
RegisterBattery	battery ID, model, chemistry, manufacturer, date	unique ID and authorized manufacturer identity
RegisterSensorConfiguration	sensor map, version, pack ID, effective date	approved configuration and model compatibility
RegisterModelVersion	model ID, checksum, training data hashes, metrics	only approved role; immutable version record
AddTelemetryHash	pack ID, time window, hash, storage URI reference	valid battery and non-overlapping batch policy
RecordSensorFault	sensor ID, class, confidence, evidence hash	severity and reviewer identity
RecordCyberIncident	attack class, affected assets, time, report hash	restricted writer and incident workflow
ApproveDeployment	model/config ID, test report, approvers	endorsement by defined organizations
UpdateLifecycleStatus	in-service, repair, second-life, recycling	valid state transition only

29. Battery Passport Connection

The EU Batteries Regulation establishes an electronic battery passport from 18 February 2027 for each EV battery, each LMT battery and industrial batteries above the defined capacity threshold. The internship should not attempt a full regulatory implementation, but the provenance layer can generate evidence for individual battery identity, model information, lifecycle events, SOH records, maintenance, model versions and data integrity.

Design principle

The blockchain record should point to authoritative evidence rather than duplicate it. Access control and privacy remain necessary because some battery-passport information is public while other information is restricted to authorities or parties with legitimate interest.

PART V

Project Definition and Work Packages

A single product concept with clear ownership for three interns

30. Recommended Project Title

Primary academic title

TrustSense-BMS: Physics-Informed Virtual Sensing, Sensor Reduction and Trusted Cybersecure Analytics for EV Battery Packs

Commercial / demonstration name

VG Marine TrustSense™ - Secure Virtual Sensing and Trusted Battery Intelligence Platform

One-line description: An AI-powered battery monitoring platform that reconstructs unmeasured thermal states, minimizes physical sensing cost, detects sensor or communication manipulation, and creates a tamper-evident lifecycle record.

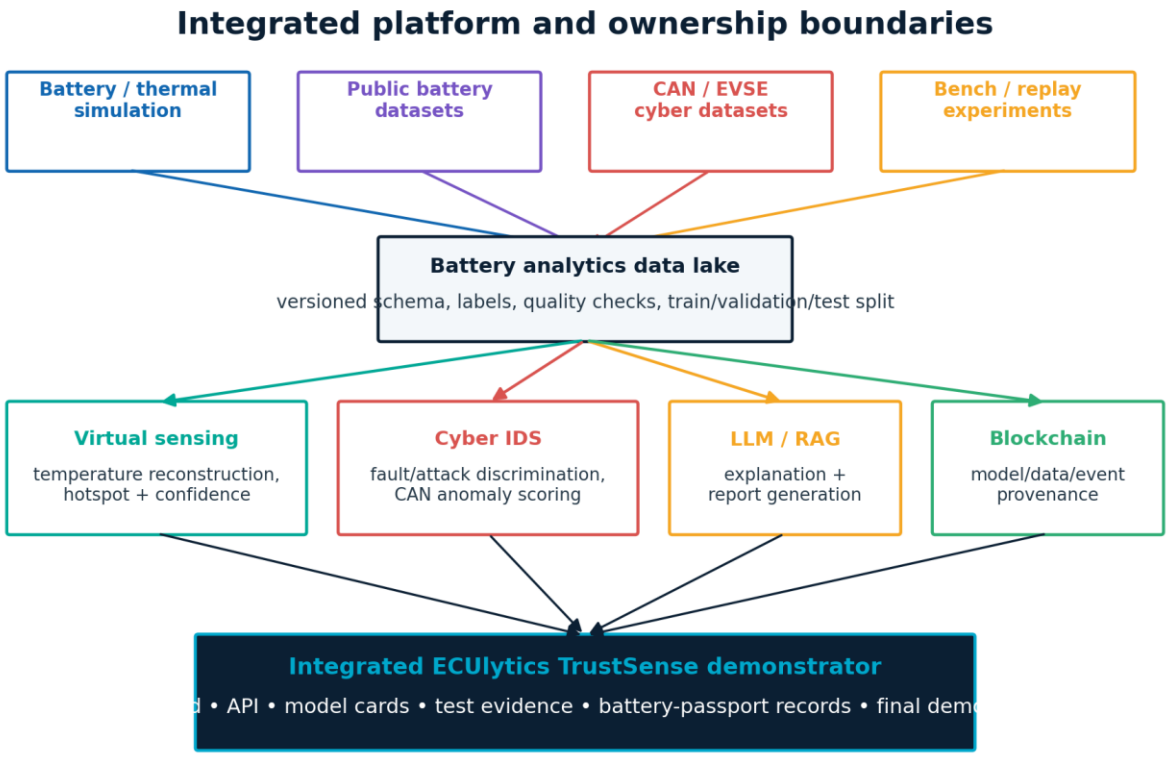


Figure 16. Integrated platform architecture and ownership boundaries.

31. Project Goals and Non-Goals

Goals	Non-goals
Reconstruct cell/module temperature fields from sparse sensing	Claim production safety certification or homologation
Quantify sensor-count versus error/hotspot/fault-isolation trade-	Remove required cell-voltage, current, isolation or contactor

Goals	Non-goals
offs	sensing
Detect sensor faults and CAN/CAN-FD manipulation	Attack a road vehicle or public charging infrastructure
Create an advisory battery-domain LLM/RAG assistant	Let an LLM directly control BMS actuators or safety thresholds
Record data/model/event provenance on a permissioned ledger	Store all telemetry on blockchain
Deliver reproducible code, tests, documentation and a demo	Present synthetic results as field-proven production performance

32. Intern 1 Work Package - Virtual Sensing and Sensor Optimization

Work-package title

Physics-Informed Virtual Temperature Sensing and Optimal Sensor Placement for EV Battery Packs

Area	Required work
Theory	ECM, heat generation, lumped thermal models, thermal networks, observability and uncertainty
Data	Normalize battery datasets; generate pack-level synthetic temperature fields and sensor masks
Baselines	Interpolation, thermal RC/Kalman filter, Random Forest or XGBoost
Advanced model	TCN/LSTM plus GNN or hybrid physics-informed model
Optimization	Uniform, random, greedy and PCA/QR or information-based sensor placement
Outputs	temperature map, maximum temperature, hotspot location, gradient, sensor-health residual and confidence
Deployment	Python API and optional edge benchmark / quantized model

Milestone	Acceptance checkpoint
M3	Data parser, thermal baseline and reproducible notebook
M5	Sensor-ablation report and selected advanced architecture
M7	Uncertainty, OOD and sensor-fault recovery integrated
M9	Frozen model and final untouched test evidence

33. Intern 2 Work Package - Cybersecure Battery Sensing

Work-package title

Cybersecure Battery Sensor Virtualization and CAN Intrusion Detection Using Time-Series AI

Area	Required work
Threat modelling	Sensor/CAN/diagnostic/cloud/AI attack surface and misuse cases
Data	HCRL, CICIoV2024, CICEVSE2024 and synthetic battery-sensor fault injection
Baselines	Rules, timing statistics, Isolation Forest and One-Class SVM
Advanced model	Autoencoder/TCN/Transformer using timing, payload and physics residuals
Classification	normal, physical fault, spoofing, replay, flooding, fuzzing, dropout and stealth bias
Outputs	anomaly score, class, affected IDs/sensors, evidence features, severity and confidence

Area	Required work
Integration	Real-time replay service and event API to blockchain/LLM

Milestone	Acceptance checkpoint
M3	Threat register, public dataset pipeline and rule baseline
M5	Unsupervised and supervised IDS benchmark
M7	Physics-aware fault/attack discrimination and open-set handling
M9	Real-time replay demonstration with incident evidence

34. Intern 3 Work Package - Blockchain and Battery Analytics LLM

Work-package title

Blockchain-Based Trusted Battery Telemetry and LLM-Assisted Battery Lifecycle Intelligence

Area	Required work
Blockchain	Hyperledger Fabric network, identities, policies, peers and chaincode
Data provenance	Hashes for raw data, feature sets, model artifacts, test reports and incidents
Smart contracts	Battery, sensor configuration, model version, telemetry, fault, incident and lifecycle events
LLM/RAG	Approved battery theory, model cards, test results, incident templates and battery-passport references
LLM safety	Role-based retrieval, citations, prompt-injection tests and advisory-only output
Outputs	verification API, audit view, report draft and battery evidence summary
Integration	Consume events from Interns 1 and 2; expose dashboard and query APIs

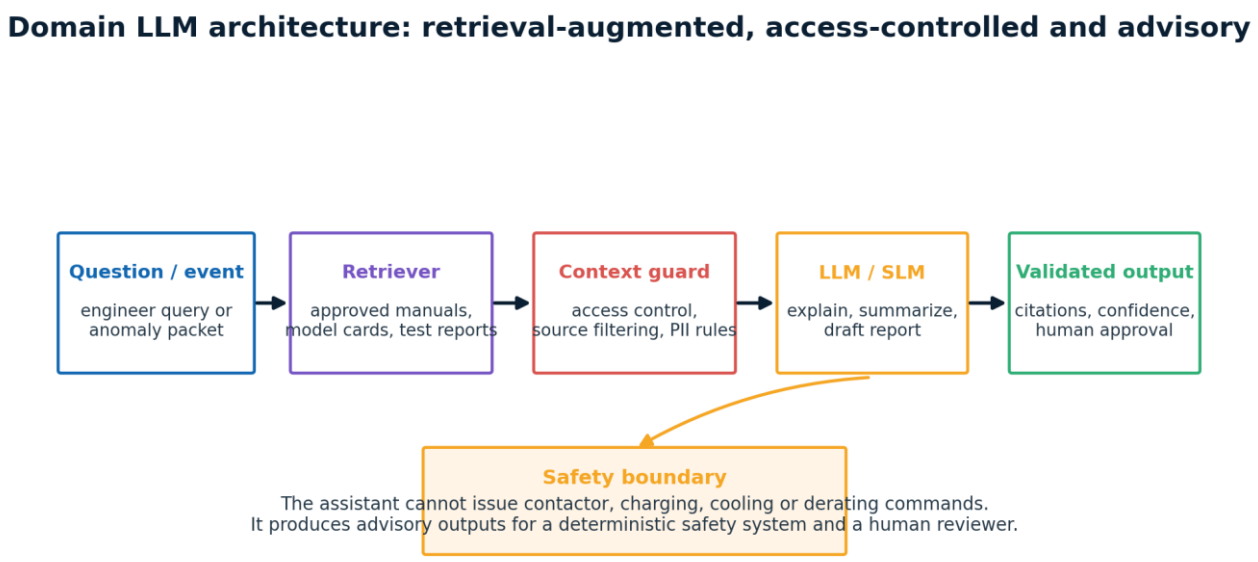


Figure 17. Recommended advisory LLM/RAG architecture.

Milestone	Acceptance checkpoint
M3	Fabric test network and basic battery/model chaincode
M5	Off-chain storage plus hash verification API

Milestone	Acceptance checkpoint
M7	RAG assistant with citations, access control and incident template
M9	Integrated provenance and passport-evidence demonstration

35. Shared Interfaces and Team Rules

Shared item	Rule
Canonical time	UTC timestamp plus explicit sampling period and sequence counter
Signal naming	battery_id/module_id/cell_id/signal/unit; no ambiguous “temp1” labels
Units	SI units internally; temperature stored with unit metadata and conversion tested
Current sign	One documented convention across all datasets and APIs
Model interface	versioned JSON/Parquet input schema and structured output with confidence
Experiment tracking	configuration, code commit, data hash, model hash and metrics captured
Safety	No live attack injection into a road vehicle; no autonomous BMS control
Review	Weekly integration review; monthly demonstration against acceptance criteria

PART VI

Data Strategy and Public Datasets

How to build credible algorithms without having a large proprietary battery dataset

36. Data Strategy Overview

No single public dataset contains all information required for pack-level temperature-field reconstruction, sensor placement, cyberattacks and blockchain provenance. The project should therefore use a layered data strategy: public cell datasets for electro-thermal dynamics, public automotive/EVSE cyber datasets for intrusion patterns, simulation for full spatial ground truth, and controlled synthetic fault injection for labels.

Layer	Purpose	Typical source
A - public battery data	Voltage/current/temperature/degradation dynamics	NASA, CALCE, Oxford, MATR, DOE Battery Data Hub, Battery Archive
B - physics simulation	Full hidden states and controllable conditions	PyBaMM plus lumped module thermal network / MATLAB / FEA if available
C - cyber data	CAN/CAN-FD and EVSE attack traffic	HCRL, CICIoV2024 and CICEVSE2024
D - synthetic fault injection	Precise labels for drift, bias, dropout and replay	Software transformation of normal battery and CAN data
E - small bench data	Reality check for timing, noise and thermal lag	Safe low-voltage emulator, thermal dummy or supervised cell/module bench

37. Recommended Public Battery Datasets

Dataset	Useful content	Primary use	Important limitation
NASA PCoE Li-ion aging / randomized use	charge, discharge, temperature, impedance and run-to-failure series	ECM, temperature trend, SOH/RUL and sequence pretraining	mostly cell-level; not a full pack spatial map
CALCE Battery Research Data	full/partial cycles, storage, dynamic profiles, OCV, impedance; multiple form factors/chemistries	cross-chemistry validation and state estimation	test protocols differ; careful metadata harmonization required
Oxford Battery Degradation Dataset 1	long-term cycling of eight pouch cells	degradation and cell-disjoint time-series validation	small cell count and one main cell type
MATR / Stanford-MIT fast-charge data	per-cycle capacity, resistance and temperature plus within-cycle traces	large-scale ML and cycle-disjoint splits	focused on fast-charge cycle-life study, not pack geometry
DOE Battery Data Hub	public advanced battery R&D datasets and tools	additional physics-informed ML sources	dataset access/metadata vary by contribution
Battery Archive	cross-institution visualization and comparison	discover and compare cycling studies	some complete files may require request

38. Recommended Public Cybersecurity Datasets

Dataset	Content	Use in project	Limitation / caution
HCRL Car-Hacking	real-vehicle CAN with DoS, fuzzing, gear and RPM spoofing	baseline CAN IDS and timing/payload features	attacks can be easy to detect; vehicle signals differ from battery CAN
HCRL CAN-FD Intrusion	real CAN-FD normal data with flooding, fuzzing and malfunction injection	CAN-FD pipeline and payload-length handling	limited duration and selected attack classes

Dataset	Content	Use in project	Limitation / caution
CICIoV2024	Ford-platform CAN with benign, DoS and signal spoofing scenarios	modern labelled IDS benchmark and representation comparison	stationary/disassembled platform conditions should be documented
CICEVSE2024	EVSE power, network traffic and host/kernel events under benign and attacks	charging-side multimodal cybersecurity extension	EVSE rather than in-pack BMS data
ROAD / curated multi-vehicle sets	stealthier fabrication/masquerade and cross-vehicle evaluation	advanced generalization benchmark	may require additional preprocessing and signal interpretation

39. Synthetic Battery Data Generation

PyBaMM is an open-source Python battery modelling framework containing electrochemical and equivalent-circuit models, parameter libraries and experiment tools. It can generate current, voltage, SOC, heat and temperature-related states for multiple operating profiles. A separate module-level thermal network should then represent cell-to-cell heat transfer and cooling geometry so that ground-truth temperatures exist for every virtual location.

Simulation dimension	Recommended values / variation
Chemistry / parameter set	LFP, NMC and NCA where supported; document the source and validity of each set
Ambient temperature	at least cold, nominal and hot ranges appropriate to the chosen cell data
Current profile	constant C-rate, pulses, drive-cycle-like dynamic profile, charge and regeneration
SOC window	low, middle, high and full usable range
Ageing proxy	increased resistance, reduced capacity and parameter variation
Cooling	normal flow, reduced flow, inlet-temperature change, fan/pump command mismatch
Cell variation	capacity, resistance and contact-conductance spread
Faults	sensor bias/drift/dropout, local heat-source increase and cooling blockage

Module node model: $m_i C_{p,i} \frac{dT_i}{dt} = Q_i - \sum_j \frac{(T_i - T_j)}{R_{ij}} - \frac{(T_i - T_{coolant})}{R_{i,cool}}$

40. Canonical Dataset Schema

Field group	Example fields
Identity	battery_id, module_id, cell_id, sensor_id, chemistry, geometry_version
Time	timestamp_utc, sequence_number, sample_period_ms, cycle_id, experiment_id
Electrical	pack_current_A, pack_voltage_V, cell_voltage_V, SOC_pct, SOH_pct, C_rate
Thermal	temperature_measured_C, temperature_ground_truth_C, ambient_C, coolant_in_C, coolant_out_C, flow
Position	row, column, xyz coordinates, cooling_distance, boundary_flag, neighbour list
Quality	validity flag, missing flag, calibration version, time-alignment quality
Fault / attack	fault_type, attack_type, start/end, affected ID, injection magnitude, label confidence
Model output	temperature_estimate_C, lower/upper interval, hotspot_probability, anomaly_score, model_version
Provenance	raw_file_hash, feature_hash, code_commit, experiment_config_hash, ledger_transaction reference

41. Data Splitting and Leakage Prevention

Time-series battery datasets are highly correlated. Randomly shuffling individual samples can place nearly identical points from the same cell and cycle in both training and test sets, producing unrealistically high scores. Splits should occur by cell, cycle, operating profile, vehicle or experiment as appropriate.

Split	Purpose	Recommended rule
Training	fit model and feature transformations	selected cells/cycles; augmentation only here
Validation	architecture, thresholds and sensor placement	different cells/cycles; never used for final reporting
Test A	in-distribution generalization	untouched cells and drive cycles from known chemistry/range
Test B	transfer / stress	unseen ambient, chemistry or geometry condition
Test C	fault/cyber robustness	unseen fault magnitude or attack intensity

Mandatory reproducibility rule

Every reported result must identify the dataset version, exact split list, preprocessing configuration, code commit, random seed and model hash. Final test data remains sealed until Month 8.

PART VII

Testing, Timeline and Acceptance Criteria

A measurable nine-month plan with evidence rather than demonstration-only claims

42. Validation Philosophy

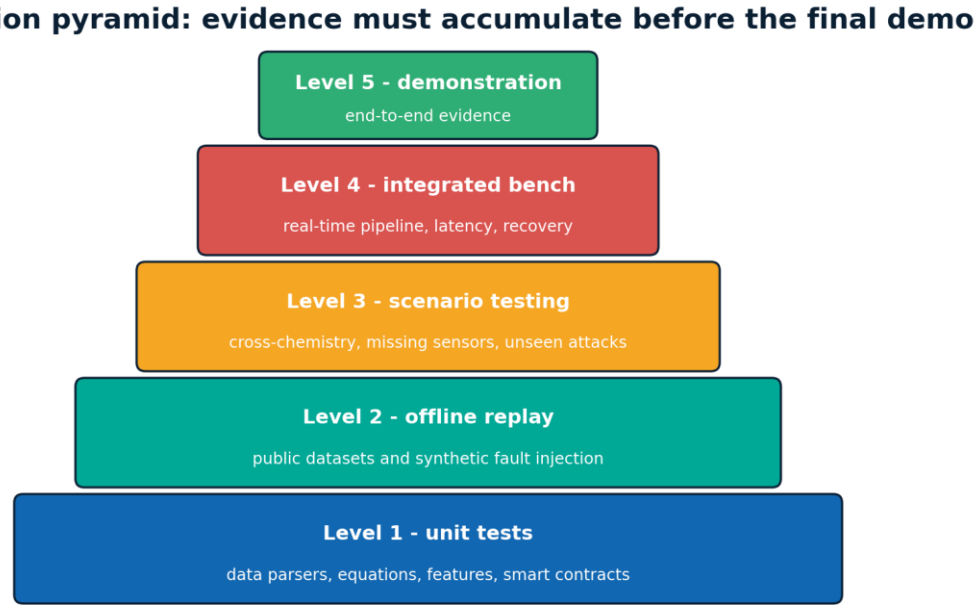


Figure 18. Validation pyramid for the integrated project.

The project should progress from deterministic unit tests to offline data replay, then stress scenarios and an integrated demonstration. A final dashboard is not sufficient evidence; every displayed result must be traceable to a repeatable test and stored configuration.

43. Virtual Sensor Test Matrix

Test	Method	Metrics / pass evidence
Normal reconstruction	Known sensors masked according to selected configuration	MAE, RMSE, max error and confidence coverage
Sensor ablation	Retain 100%, 75%, 50%, 37.5%, 25% and lower where feasible	trade-off curve and selected operating point
Hotspot localization	Introduce or simulate local heating	hotspot recall, location error and detection delay
Unseen drive cycle	Hold out complete current profile	generalization without retraining
Cross-temperature	Train and test at separated ambient ranges	error degradation and OOD behaviour
Cross-chemistry / transfer	Hold out chemistry or parameter set	transfer result; recalibrate if required
Sensor fault	bias, drift, stuck, dropout and time shift	recovery error, fault detection delay, false alarm
Cooling fault	reduced flow / altered boundary condition	hotspot and uncertainty response

Test	Method	Metrics / pass evidence
Real-time	stream input at target sample rate	latency percentile, CPU and memory

44. Cybersecurity Test Matrix

Test	Method	Metrics / pass evidence
Benign false alarms	long normal replay across datasets	false-positive rate and false alarms per hour
Known attack classification	labelled DoS/fuzz/spoof/replay windows	precision, recall, F1, PR-AUC and delay
Unseen intensity	different injection rate or bias magnitude	robustness curve and open-set response
Stealth bias	slow in-range sensor manipulation	physics-aware detection performance
Cross-vehicle / cross-dataset	train on one source, test on another	generalization score and calibration
Combined fault and attack	physical fault plus CAN anomaly	multi-label/root-cause evidence
Adversarial ML	input perturbation or poisoned subset in controlled environment	degradation, detection and mitigation evidence
Recovery	stop attack / restore sensor	return-to-normal time and retained event record

45. Blockchain and LLM Test Matrix

Test	Method	Pass evidence
Hash verification	modify one byte in an off-chain file	verification fails and identifies affected batch
Unauthorized transaction	use identity without required role	transaction rejected by policy
Endorsement	omit required organization approval	deployment/event transaction rejected
Peer interruption	stop and restore one peer	network recovers according to configured architecture
Version trace	query model used for a selected estimate	model, data and test evidence returned
RAG citation	ask battery/cyber question	response identifies approved source and does not fabricate unsupported control action
Prompt injection	place malicious instruction in untrusted document	retrieval/context guard blocks or labels it
Access control	restricted document query by unauthorized role	content not retrieved or exposed

46. Development Targets

Targets are project gates, not production guarantees

The numerical targets below should be adjusted after Month 2 based on available data, sensor type and simulated geometry. They are intended to prevent vague success criteria; achieving them does not demonstrate automotive safety compliance.

Area	Initial target
Virtual temperature normal-condition RMSE	<= 1.0 °C on the defined in-distribution test set
Maximum absolute error	<= 2.0 °C for normal conditions, with exceptions documented
Hotspot recall	>= 95% for defined hotspot threshold/scenarios

Area	Initial target
Sensor reduction	demonstrate at least 50% reduction versus full instrumentation while meeting agreed criteria
Uncertainty coverage	90% interval contains approximately 90% of test targets within tolerance
CAN IDS known-attack F1	>= 0.95 on the primary dataset and transparent split
Cross-dataset / transfer F1	>= 0.80-0.85 or documented reason and improvement plan
False alarms	<= 1% sample/window FPR plus false alarms per hour reported
High-rate attack detection delay	<= 500 ms in the replay environment
Model/API latency	p95 <= 100 ms at chosen sampling/load, or justified architecture
Blockchain integrity	100% detection of deliberately altered referenced files
Reproducibility	fresh setup reproduces key metrics within stated tolerance

47. Nine-Month Roadmap

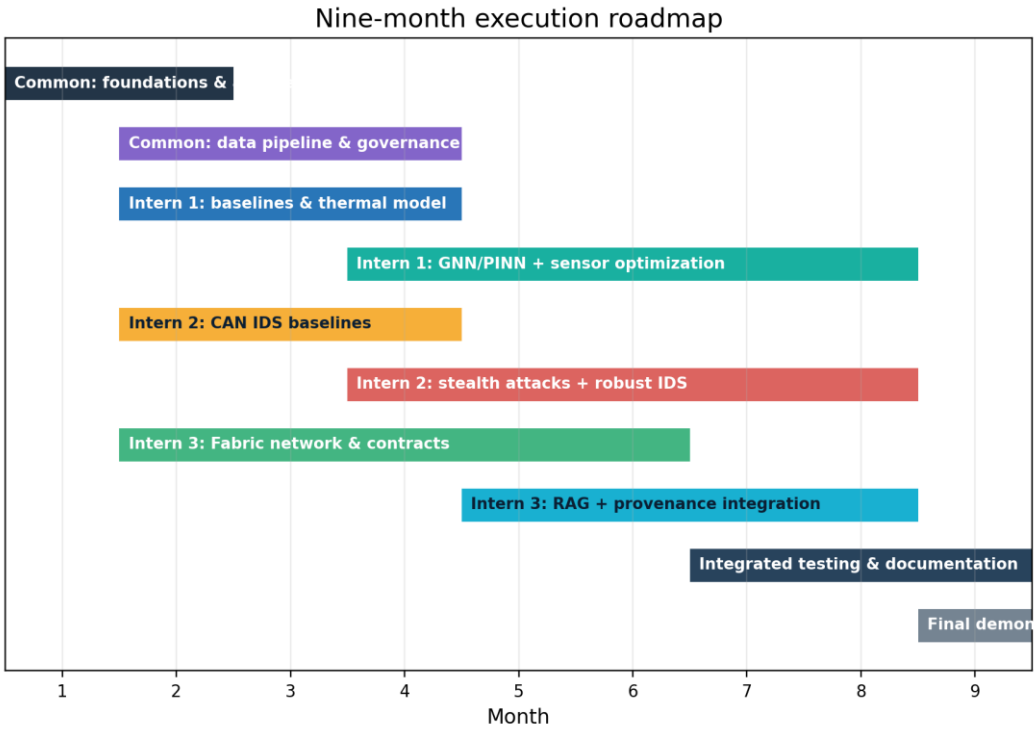


Figure 19. Nine-month workstream roadmap.

Month	Common team output	Intern 1	Intern 2	Intern 3
1	requirements, learning tests, architecture and repository	battery/thermal theory and schema	threat model and CAN basics	Fabric/RAG fundamentals and governance
2	public data ingestion and data-quality report	ECM/thermal simulator baseline	cyber dataset pipeline and timing rules	Fabric test network and identity plan
3	first integrated data contract	Kalman/tree baseline	Isolation Forest/autoencoder baseline	battery/model smart contracts
4	mid-term benchmark report	TCN/LSTM/GNN prototype	supervised IDS and physical residuals	off-chain storage and hash API
5	sensor/attack ablation review	sensor placement comparison	stealth fault/attack experiments	model approval and event contracts
6	integration sprint 1	uncertainty and OOD	open-set and cross-dataset test	RAG assistant with approved sources
7	integration sprint 2	real-time API and sensor recovery	real-time replay and incident API	dashboard/provenance integration
8	frozen final-test release	sealed test and transfer study	sealed attack/cross-dataset test	security, tamper and prompt-injection test
9	final demo, report, code handover	model card and sensor-reduction	cybersecurity report and playbook	ledger/RAG deployment guide and

Month	Common team output	Intern 1	Intern 2	Intern 3
	and viva	report		audit report

48. Weekly Operating Rhythm

Cadence	Activity	Evidence
Monday	30-minute planning and dependency review	weekly issue list and measurable goals (internal)
Mid-week	pair integration or technical review	merged interface test or reviewed notebook (internal)
Friday	demonstration and metric review	chart, test log and updated experiment tracker
Monthly	formal milestone gate	signed acceptance checklist and risk update (with VG)

49. Final Deliverables

- Battery/thermal theory notebook and parameter glossary.
- Dataset ingestion, validation and canonical-schema package.
- PyBaMM experiment generator and module thermal-network simulator.
- Virtual temperature model, uncertainty layer and sensor-placement optimizer.
- Sensor-fault injector and fault-recovery evaluation suite.
- CAN/CAN-FD replay pipeline and intrusion-detection service.
- Threat model, attack catalogue and cybersecurity test report.
- Hyperledger Fabric network configuration, chaincode and verification API.
- Battery-domain RAG assistant, approved knowledge base and prompt-injection tests.
- Integrated web dashboard/API with traceable experiment identifiers.
- Model cards, data cards, installation guide, user guide and final presentation.
- Final demonstration recording and reproducibility package.

50. Evaluation Rubric

Category	Weight	What earns full credit
Technical correctness	20%	physics, units, sign conventions and algorithms are correct and reviewed
Data discipline	15%	lineage, leakage prevention, quality checks and reproducible splits
Model evidence	20%	baselines, untouched tests, uncertainty and transparent limitations
Cybersecurity	15%	realistic threat model, safe test method and robust detection evidence
Integration	10%	stable interfaces, end-to-end workflow and provenance
Software quality	10%	tests, typing, logging, configuration and deployment documentation
Communication	10%	clear theory explanation, visualizations, model/data cards and viva

APPENDICES

Implementation Aids and References

Glossary, folder structure, checklists and primary public sources

Appendix A - Suggested Repository Structure

```
trustsense-bms/
  docs/
    handbook/  architecture/  threat-model/  model-cards/  data-cards/
  configs/
    datasets/  experiments/  models/  fabric/
  data/
    raw/  interim/  processed/  synthetic/  splits/
  src/
    common/  battery_models/  virtual_sensors/  cyber_ids/  blockchain/  rag/
    api/
  tests/
    unit/  integration/  replay/  acceptance/
  notebooks/
    learning/  exploration/  baselines/
  deployment/
    docker/  edge/  fabric-network/
  reports/
    metrics/  plots/  incidents/  final/
  README.md  pyproject.toml  LICENSE  SECURITY.md
```

Example only; the team may adapt it while preserving separation of raw data, code, configuration and evidence.

Appendix B - Minimum Model Card

Field	Required content
Model identity	name, semantic version, checksum and code commit
Intended use	specific battery, chemistry, geometry, signals and advisory/safety boundary
Training data	dataset names, hashes, split IDs and synthetic-data configuration
Inputs/outputs	names, units, sampling, range, missing-data behavior and confidence
Metrics	normal, stress, fault, transfer and latency results with confidence intervals
Limitations	unsupported chemistry/range/geometry, known failure modes and OOD behaviour
Security	poisoning/evasion considerations, signed artifact and access control
Approval	reviewer, date, test report hash and blockchain record

Appendix C - Glossary

Term	Definition
BMS	Battery Management System
CAN	Controller Area Network
CAN-FD	CAN with Flexible Data Rate
Cell	Smallest electrochemical unit in the pack
C-rate	Current normalized by nominal capacity
ECM	Equivalent Circuit Model
GNN	Graph Neural Network
Hotspot	Location with maximum or risk-relevant temperature
LLM	Large Language Model
MAE	Mean Absolute Error
OOD	Out of Distribution
OCV	Open-Circuit Voltage
PINN	Physics-Informed Neural Network
RAG	Retrieval-Augmented Generation
RMSE	Root Mean Square Error
RUL	Remaining Useful Life
SOC	State of Charge
SOH	State of Health
Virtual sensor	Software estimate replacing or checking an unavailable physical measurement
Analytical redundancy	Model/relationship that checks a measurement without duplicate hardware

Appendix D - Primary Online Data and Standards Sources

The following sources were selected because they are official repositories, standards pages, primary documentation or original research records. Dataset licenses and citation requirements must be checked before redistribution or commercial use.

[1] NASA Prognostics Center of Excellence Data Set Repository. Source link

[2] NASA Open Data: Li-ion Battery Aging Datasets. Source link

[3] CALCE Battery Research Data. Source link

[4] Oxford Battery Degradation Dataset 1. Source link

[5] MATR / Stanford-MIT fast-charge dataset code and data description. Source link

[6] U.S. DOE Battery Data Hub. Source link

[7] Battery Archive. Source link

[8] PyBaMM documentation. Source link

[9] HCRL Car-Hacking Dataset. Source link

[10] HCRL CAN-FD Intrusion Dataset. Source link

[11] CICIoV2024. Source link

[12] CICEVSE2024. Source link

[13] ISO/SAE 21434:2021 official overview. Source link

[14] NISTIR 8473: Cybersecurity Framework Profile for EV/XFC Infrastructure. Source link

[15] NIST AI 100-2e2025: Adversarial Machine Learning taxonomy. Source link

[16] NIST AI 600-1: Generative AI Profile. Source link

[17] Hyperledger Fabric documentation. Source link

[18] EU Regulation 2023/1542 on batteries and battery passport. Source link

Appendix E - Selected Research Reading

Reference	Title	Publication
Fang et al. (2021)	Temperature-Field Sparse-Reconstruction of Lithium-Ion Battery Pack Based on Artificial Neural Network and Virtual Thermal Sensor Technology	Energy Technology, DOI: 10.1002/ente.202100258.
Zhao et al. (2024)	Temperature field spatiotemporal modeling of lithium-ion battery pack configured sparse temperature sensors	Journal of Energy Storage 97, 113005, DOI: 10.1016/j.est.2024.113005.
Cheng, D'Arpino and Rizzoni (2020)	Optimal Sensor Placement in Lithium-Ion Battery Pack for Fault Detection and Isolation	arXiv:2008.10533.
Song, Woo and Kim (2020)	In-vehicle network intrusion detection using deep convolutional neural network	Vehicular Communications 21, 100198.
Severson et al. (2019)	Data-driven prediction of battery cycle life before capacity degradation	Nature Energy 4, 383-391.

Appendix F - Mentor Sign-Off Checklist

- ☐ All interns passed the battery fundamentals quiz and unit/sign-convention review.
- ☐ Raw datasets are immutable and documented with licenses/citations.
- ☐ Training, validation and final test splits are separated by cell/cycle/vehicle as appropriate.
- ☐ Every advanced model is compared with transparent baselines.
- ☐ Sensor-reduction recommendation includes hotspot and fault-isolation evidence, not only average RMSE.
- ☐ Uncertainty/OOD and fallback behaviour are tested.
- ☐ Cyberattack experiments are performed only in safe offline, virtual or isolated test environments.
- ☐ LLM outputs remain advisory and include source evidence.
- ☐ Blockchain stores hashes/events rather than all high-rate telemetry.
- ☐ Final code, configuration, model/data cards and tests reproduce on a clean system.

Closing Note

The strongest outcome of this internship will not be a single high accuracy number. It will be a disciplined, explainable and reproducible battery analytics platform that demonstrates how software can reduce sensing cost while preserving engineering evidence, cyber resilience and data trust. The interns should finish with an understanding that battery AI is a cyber-physical engineering discipline: every model output has units, physical assumptions, uncertainty, failure modes and a safety boundary.

VG Marine project outcome

A customer-demonstrable foundation for secure battery intelligence: virtual sensing, thermal analytics, anomaly detection, model/data provenance and battery-passport evidence - developed with clear limitations and a roadmap toward controlled industrial validation.